

MH1101 Tutorial 11 (Week 12)

Solution

Reference: Topic 6.1, 6.2 (Lecture Notes)

1. Find the radius of convergence and interval of convergence of the series.

$$(a) \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)2^n} (x-1)^n$$

$$(b) \sum_{n=1}^{\infty} \frac{\sqrt{n}}{8^n} (x+6)^n$$

$$(c) \sum_{n=2}^{\infty} \frac{b^n}{\ln n} (x-a)^n, b > 0$$

$$(d) \sum_{n=1}^{\infty} \frac{x^n}{1 \cdot 3 \cdot 5 \cdots (2n-1)}$$

Solution

(a) Let $a_n = \frac{(-1)^n}{(2n-1)2^n} (x-1)^n$. Then

$$\begin{aligned} \rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1}}{(2(n+1)-1)2^{n+1}} (x-1)^{n+1} \cdot \frac{(2n-1)2^n}{(-1)^n (x-1)^n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{2n-1}{2n+1} \cdot \frac{|x-1|}{2} = \frac{|x-1|}{2}. \end{aligned}$$

By the Ratio Test, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)2^n} (x-1)^n$ converges when $\frac{|x-1|}{2} < 1 \iff |x-1| < 2$ (so $R = 2$) $\iff -2 < x-1 < 2 \iff -1 < x < 3$.

(i) When $x = 3$, the series becomes

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)2^n} (x-1)^n = \sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{(2n-1)2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)}$$

which converges by the Alternating series test, since $\left\{\frac{1}{2n-1}\right\}$ is decreasing to 0.

(ii) When $x = -1$, the series becomes

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)2^n} (x-1)^n = \sum_{n=1}^{\infty} \frac{(-1)^n (-2)^n}{(2n-1)2^n} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)}$$

which diverges by the limit Comparison Test with the Harmonic series $\sum \frac{1}{n}$:

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{2n-1}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n}{2n-1} = \frac{1}{2} > 0.$$

Hence, the interval of convergence is $(-1, 3]$.

(b) Let $a_n = \frac{\sqrt{n}}{8^n} (x+6)^n$. Then

$$\begin{aligned} \rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{\sqrt{n+1}}{8^{n+1}} (x+6)^{n+1} \cdot \frac{8^n}{\sqrt{n} (x+6)^n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{1}{8} \sqrt{\frac{n+1}{n}} |x+6| \\ &= \frac{|x+6|}{8}. \end{aligned}$$

By the Ratio Test, the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{8^n} (x+6)^n$ converges when $\frac{|x+6|}{8} < 1$
 $\iff |x+6| < 8$ (so $R = 8$) $\iff -8 < x+6 < 8 \iff -14 < x < 2$.

(i) When $x = 2$, the series becomes

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{8^n} (x+6)^n = \sum_{n=1}^{\infty} \frac{\sqrt{n} 8^n}{8^n} = \sum_{n=1}^{\infty} \sqrt{n}$$

which diverges by the n -th Term Test for Divergence.

(ii) When $x = -14$, the series becomes

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{8^n} (x+6)^n = \sum_{n=1}^{\infty} \frac{\sqrt{n} (-8)^n}{8^n} = \sum_{n=1}^{\infty} (-1)^n \sqrt{n}$$

which diverges by the n -th Term Test for Divergence.

Hence, the interval of convergence is $(-14, 2)$.

(c) Let $a_n = \frac{b^n}{\ln n}(x - a)^n$. Then

$$\begin{aligned}\rho &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{b^{n+1}(x - a)^{n+1}}{\ln(n+1)} \cdot \frac{\ln n}{b^n(x - a)^n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{\ln n}{\ln(n+1)} \cdot b|x - a| \\ &= \lim_{n \rightarrow \infty} \frac{1/n}{1/(n+1)} \cdot b|x - a| \quad (\text{by L'Hopital Rule}) \\ &= b|x - a|.\end{aligned}$$

By the Ratio Test, the series $\sum_{n=2}^{\infty} \frac{b^n}{\ln n}(x - a)^n$ converges when $b|x - a| < 1$
 $\iff |x - a| < \frac{1}{b}$ (so $R = \frac{1}{b}$) $\iff -\frac{1}{b} < x - a < \frac{1}{b} \iff a - \frac{1}{b} < x < a + \frac{1}{b}$.

(i) When $x = a + \frac{1}{b}$, the series becomes

$$\sum_{n=2}^{\infty} \frac{b^n}{\ln n}(x - a)^n = \sum_{n=2}^{\infty} \frac{b^n}{\ln n} \frac{1}{b^n} = \sum_{n=2}^{\infty} \frac{1}{\ln n}$$

which diverges by Comparison Test: Since

$$\frac{1}{\ln n} > \frac{1}{n}$$

the series $\sum_{n=2}^{\infty} \frac{1}{\ln n}$ diverges as the Harmonic series $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges.

(ii) When $x = a - \frac{1}{b}$, the series becomes

$$\sum_{n=2}^{\infty} \frac{b^n}{\ln n}(x - a)^n = \sum_{n=2}^{\infty} \frac{b^n}{\ln n} \frac{(-1)^n}{b^n} = \sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$$

which converges by the Alternating Series Test, since the sequence $\left\{ \frac{1}{\ln n} \right\}$ is decreasing to 0.

Hence, the interval of convergence is $\left[a - \frac{1}{b}, a + \frac{1}{b} \right)$.

(d) Let $a_n = \frac{x^n}{1 \cdot 3 \cdot 5 \cdots (2n-1)}$. Then

$$\begin{aligned}\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{x^{n+1}}{1 \cdot 3 \cdot 5 \cdots (2n-1)(2(n+1)-1)} \cdot \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{x^n} \\ &= \lim_{n \rightarrow \infty} \frac{|x|}{2n+1} = 0.\end{aligned}$$

The series converges for all x , i.e. $R = \infty$, and the interval of convergence is $(-\infty, \infty)$.

□

2. Suppose that the power series $\sum c_n(x-a)^n$ satisfies $c_n \neq 0$ for all n .

Show that if $\lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$ exists, then it is equal to the radius of convergence of the power series.

Solution Let

$$\lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right| = L \in \mathbb{R}.$$

Consider two cases: $L = 0$ and $L > 0$.

- $L > 0$.

Let $a_n = c_n(x-a)^n$. Since $c_n \neq 0$, applying the Ratio Test to the series $\sum a_n$, we have

$$\begin{aligned}\rho &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n} \right| \\ &= \left(\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| \right) \cdot |x-a| \\ &= \frac{1}{\lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|} \cdot |x-a| \\ &= \frac{1}{L} |x-a|.\end{aligned}$$

The series converges when $\frac{1}{L}|x - a| < 1$, i.e. $|x - a| < L$ where L is the radius of convergence.

- $L = 0$.

If $x \neq a$, then $\rho = |x - a| \lim_{n \rightarrow \infty} \left| \frac{1}{\frac{c_n}{c_{n+1}}} \right| = \infty$ (since $\left| \frac{c_n}{c_{n+1}} \right| \rightarrow 0$ and $|x - a| \neq 0$), and so by the Ratio Test, the series diverges for all $x \neq a$. Clearly, the power series centered at a is always converges at $x = a$. So in this case, the radius of convergence is $R = 0 = L$.

Hence, in all cases, we have proved that the power series has radius of convergence equal to $\lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$. $\square$

3. Suppose the series $\sum c_n x^n$ has radius convergence R . What is the radius of convergence of the power series $\sum c_n x^{2n}$? Justify your answer.

Solution The series $\sum c_n x^n$ converges whenever $|x| < R$. Since

$$\sum c_n x^{2n} = \sum c_n (x^2)^n,$$

the series $\sum c_n x^{2n}$ converges whenever $|x^2| < R \iff |x| < \sqrt{R}$, and so its radius of convergence is $\sqrt{R}$. $\square$

4. Find a power series (in x) representation for the function by manipulating a geometric series and determine the interval of convergence.

(a) $f(x) = \frac{4}{2x + 3}$

(b) $f(x) = \frac{x^2}{x^4 + 16}$

(c) $f(x) = \frac{x - 1}{x + 2}$

Solution

(a)

$$\begin{aligned}f(x) &= \frac{4}{2x+3} = \frac{4}{3} \cdot \frac{1}{1+(2x/3)} \\&= \frac{4}{3} \cdot \frac{1}{1-(-2x/3)} \\&= \frac{4}{3} \sum_{n=0}^{\infty} \left(-\frac{2x}{3}\right)^n \quad \text{or equivalently} \\&= \sum_{n=0}^{\infty} (-1)^n \frac{2^{n+2}}{3^{n+1}} x^n\end{aligned}$$

The series converges if and only if $\left|-\frac{2x}{3}\right| < 1 \iff$

$$|x| < \frac{3}{2},$$

and so $R = \frac{3}{2}$, and the interval convergence I is $(-\frac{3}{2}, \frac{3}{2})$.

(b)

$$\begin{aligned}f(x) &= \frac{x^2}{x^4+16} \\&= \frac{x^2}{16} \cdot \frac{1}{1+(x^4/16)} \\&= \frac{x^2}{16} \cdot \frac{1}{1-(-x^4/16)} \\&= \frac{x^2}{16} \sum_{n=0}^{\infty} \left(-\frac{x^4}{16}\right)^n \quad \text{or equivalently} \\&= \sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+2}}{2^{4n+4}}.\end{aligned}$$

The series converges if and only if $\left|-\frac{x^4}{16}\right| < 1 \iff$

$$|x| < 16^{1/4} = 2,$$

and so $R = 2$, and the interval convergence I is $(-2, 2)$.

(c)

$$\begin{aligned}f(x) &= \frac{x-1}{x+2} \\&= 1 - \frac{3}{x+2} \\&= 1 - \frac{3}{2} \cdot \frac{1}{1+(x/2)} \\&= 1 - \frac{3}{2} \cdot \frac{1}{1-(-x/2)} \\&= 1 - \frac{3}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n \\&= 1 - \frac{3}{2} - \frac{3}{2} \sum_{n=1}^{\infty} \left(-\frac{x}{2}\right)^n \\&= -\frac{1}{2} - \sum_{n=1}^{\infty} (-1)^n \frac{3x^n}{2^{n+1}}.\end{aligned}$$

The series converges if and only if $\left|-\frac{x}{2}\right| < 1 \iff$

$$|x| < 2,$$

and so $R = 2$, and the interval convergence I is $(-2, 2)$.

□

5. By differentiating or integrating certain power series, find a power series (in x) representation for the function and determine the radius of convergence.

(a) $f(x) = \ln(5-x)$

(b) $f(x) = \left(\frac{x}{2-x}\right)^3$

Solution

(a) Recall that $\int \frac{1}{x} dx = \ln |x|$. Thus, for $5 - x > 0$, we have

$$\ln(5 - x) = - \int \frac{1}{5 - x} dx.$$

Idea: expanding $\frac{1}{5-x}$ using geometric series, and integrating term-by-term, we will obtain a series representation of $\ln(5 - x)$.

Note that

$$\begin{aligned} \frac{1}{5 - x} &= \frac{1}{5} \cdot \frac{1}{1 - (x/5)} \\ &= \frac{1}{5} \sum_{n=0}^{\infty} \left(\frac{x}{5}\right)^n \end{aligned}$$

which converges if and only if $|x/5| < 1$ i.e. $|x| < 5$, and so the radius of convergence R is 5.

Hence,

$$\begin{aligned} \ln(5 - x) &= - \int \frac{1}{5 - x} dx \\ &= - \int \frac{1}{5} \sum_{n=0}^{\infty} \left(\frac{x}{5}\right)^n dx \\ &= - \frac{1}{5} \sum_{n=0}^{\infty} \frac{x^{n+1}}{5^n(n+1)} + C \\ &= C - \sum_{n=0}^{\infty} \frac{x^{n+1}}{5^{n+1}(n+1)} \\ &= C - \sum_{n=1}^{\infty} \frac{x^n}{5^n n} \quad (\text{after re-indexing}). \end{aligned}$$

Substituting $x = 0$, we have $C = \ln 5$. So

$$\ln(5 - x) = \ln 5 - \sum_{n=1}^{\infty} \frac{x^n}{5^n n}.$$

The series converges for $|x| < 5$, and its radius of convergence R is 5.

(b) Note that

$$\frac{d}{dx} \left(\frac{1}{2-x} \right) = \frac{1}{(2-x)^2}, \quad \frac{d^2}{dx^2} \left(\frac{1}{2-x} \right) = \frac{d}{dx} \left(\frac{1}{(2-x)^2} \right) = \frac{2}{(2-x)^3}.$$

Thus,

$$\begin{aligned} \left(\frac{x}{2-x} \right)^3 &= \frac{x^3}{2} \frac{d^2}{dx^2} \left(\frac{1}{2-x} \right) \\ &= \frac{x^3}{2} \frac{d}{dx} \left(\frac{d}{dx} \left(\frac{1}{2-x} \right) \right) \\ &= \frac{x^3}{2} \frac{d}{dx} \left(\frac{d}{dx} \left(\frac{1}{2} \cdot \frac{1}{1-(x/2)} \right) \right) \\ &= \frac{x^3}{4} \frac{d}{dx} \left(\frac{d}{dx} \left(\sum_{n=0}^{\infty} \left(\frac{x}{2} \right)^n \right) \right) \\ &= \frac{x^3}{4} \frac{d}{dx} \left(\sum_{n=0}^{\infty} \frac{d}{dx} \left(\frac{x}{2} \right)^n \right) \\ &= \frac{x^3}{4} \frac{d}{dx} \left(\sum_{n=1}^{\infty} n \left(\frac{x}{2} \right)^{n-1} \frac{1}{2} \right) \\ &= \frac{x^3}{8} \frac{d}{dx} \left(\sum_{n=1}^{\infty} n \left(\frac{x}{2} \right)^{n-1} \right) \\ &= \frac{x^3}{8} \sum_{n=1}^{\infty} \frac{d}{dx} \left(n \left(\frac{x}{2} \right)^{n-1} \right) \\ &= \frac{x^3}{2^3} \sum_{n=2}^{\infty} n(n-1) \left(\frac{x}{2} \right)^{n-2} \frac{1}{2} \\ &= \sum_{n=2}^{\infty} n(n-1) \frac{x^{n+1}}{2^{n+2}} \quad (\text{or equivalently}) \\ &= \sum_{n=0}^{\infty} (n+2)(n+1) \frac{x^{n+3}}{2^{n+4}} \quad (\text{after re-indexing}). \end{aligned}$$

The series converges for $|x/2| < 1$, i.e. $|x| < 2$, so its radius of convergence is $R = 2$.

□

6. (Some applications of power series)(a) Use the first three terms of a power series to evaluate $\int_0^{0.4} \ln(1+x^4) dx$.

(b) Find the sum of the series

$$\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n}.$$

Hint: Evaluate $\sum_{n=2}^{\infty} n(n-1)x^n$, $|x| < 1$, using a power series. Recall that in Example 6.9 of Chapter 6, we have $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$, $|x| < 1$.

Solution(a) Note that for $1+x > 0$,

$$\begin{aligned} \ln(1+x) &= \int \frac{1}{1+x} dx = \int \frac{1}{1-(-x)} dx \\ &= \int \sum_{n=0}^{\infty} (-1)^n x^n dx \\ &= \sum_{n=0}^{\infty} \int (-1)^n x^n dx \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}. \end{aligned}$$

Thus,

$$\begin{aligned} \ln(1+x^4) &= \sum_{n=0}^{\infty} \frac{(-1)^n (x^4)^{n+1}}{n+1} \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+4}}{n+1}. \end{aligned}$$

$$\begin{aligned}
\int_0^{0.4} \ln(1+x^4) dx &\approx \int_0^{0.4} \frac{x^4}{1} - \frac{x^8}{2} + \frac{x^{12}}{3} dx \\
&= \left[\frac{x^5}{5} - \frac{x^9}{18} + \frac{x^{13}}{39} \right]_0^{0.4} \\
&\approx 0.002034.
\end{aligned}$$

(b) Recall that in Example 6.9 of Chapter 6, we have $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$, $|x| < 1$. Thus,

$$\begin{aligned}
\frac{2}{(1-x)^3} &= \frac{d}{dx} \left(\frac{1}{(1-x)^2} \right) = \frac{d}{dx} \left(\sum_{n=1}^{\infty} nx^{n-1} \right) \\
&= \sum_{n=1}^{\infty} \frac{d}{dx} (nx^{n-1}) \\
&= \sum_{n=2}^{\infty} n(n-1)x^{n-2}
\end{aligned}$$

converges for $|x| < 1$.

Substituting $x = \frac{1}{2}$, we have

$$\begin{aligned}
\frac{2}{(1 - (1/2))^3} &= \sum_{n=2}^{\infty} \frac{n(n-1)}{2^{n-2}} \\
\frac{2}{(1 - (1/2))^3} &= \frac{1}{2^{-2}} \sum_{n=2}^{\infty} \frac{n(n-1)}{2^n} \\
\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n} &= 2^{-2} \cdot \frac{2}{(1 - (1/2))^3} = 4.
\end{aligned}$$

□

Answer Keys. 1. (a) $(-1, 3]$ (b) $(-14, 2)$ (c) $[a - \frac{1}{b}, a + \frac{1}{b})$ (d) $(-\infty, \infty)$

3. $\sqrt{R}$ 4. (a) $\sum_{n=0}^{\infty} (-1)^n \frac{2^{n+2}}{3^{n+1}} x^n$ (b) $\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+2}}{2^{4n+4}}$ (c) $-\frac{1}{2} - \sum_{n=1}^{\infty} (-1)^n \frac{3x^n}{2^{n+1}}$

5. (a) $C - \sum_{n=1}^{\infty} \frac{x^n}{5^n n}$ (b) $\sum_{n=0}^{\infty} (n+2)(n+1) \frac{x^{n+3}}{2^{n+4}}$ 6. (a) 0.002034 (b) 4